Yingjing Huang

Department of Geography and Regional Research
University of Vienna
Vienna, Austria

yingjing.huang@univie.ac.at
yingjinghuang.github.io
github.com/yingjinghuang
Google Scholar
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RESEARCH INTERESTS

GIScience, GeoAI, deep learning, computer vision, urban visual intelligence, social sensing

EDUCATION

Ph.D.	Cartography and Geographic Information System, Peking University, Beijing, China, 2021–2025 Advisors: Yu Liu, Fan Zhang
M.S.	Cartography and Geographic Information System, Wuhan University, Wuhan, China, 2018–2021 Advisor: Teng Fei
B.Sc.	Geographic Science, Wuhan University, Wuhan, China, 2013–2017 Advisors: Yiyun Chen, Teng Fei

EXPERIENCES

2026–	Postdoctoral Researcher, University of Vienna, Vienna, Austria Advisor: Krzysztof Janowicz
2023–2024	Visiting Ph.D. Student, MIT Senseable City Lab, Cambridge, USA Advisors: Carlo Ratti, Fábio Duarte, Simone Mora

PUBLICATIONS

Bold indicates the author; underline indicates co-first authorship.
Total citations (Google Scholar): 319.

Manuscripts under Review and in Preparation

2026	Huang, Y., Wang, Y., Hou, C., Yao, Y., Duarte, F., & Zhang, F. (2026). Measuring urban inequality across private and public spaces [Under review]. *Cities*
2026	Wang, Y., Zhang, F., Hou, C., Huang, Y., Liang, Y., Biljecki, F., Xu, L., & Zhang, L. (2026). Uncovering historical city connections with satellite imagery [Under review]. *Nature Cities*
2026	Li, J., Huang, Y., Zhang, F., & Liu, Y. (2026a). StreetTree: A large-scale global benchmark for fine-grained tree species classification [Under review; arXiv:2602.19123]. *Earth System Science Data*

2026 Li, J., Huang, Y., Zhang, F., & Liu, Y. (2026b). Global urban tree species classification through geo-aware knowledge distillation [Under review]. *Urban Forestry & Urban Greening*

2026 Wang, Y., Huang, Y., & Zhang, F. (2026). UrbanHue: Visualizing the continuous structure of high-dimensional urban features in a perceptual color space [Under review]. *International Journal of Geographical Information Science*

Peer-reviewed Journal Articles

2026 Huang, Y., & Zhang, F. (2026). Advancing urban sustainability with visual intelligence. *Science Bulletin*

2026 Li, Y., Huang, Y., & Zhang, F. (2026). Learning street view representations based on a spatiotemporal contrastive learning framework. *Computers, Environment and Urban Systems*, *125*, 102393

2025 Xiao, J., Fei, T., Yu, B., Huang, Y., & Du, Y. (2025). Street view images help to reveal the impact of noisy environments on the survival duration of stroke patients. *International Journal of Health Geographics*, *24*(1), 27

2025 Liu, Y., Wang, X., Wang, Y., Huang, F., Huang, Y., Li, Y., Zhang, W., Gong, S., Mai, G., Yao, Y., Yue, Y., Li, H., & Zhang, F. (2025). Representation learning for geospatial data. *Annals of GIS*, 1–27

2025 Zhu, J., Huang, Y., Cao, Z., Zhang, Y., Ding, Y., & Du, J. (2025). Evaluating urban greenery through the front-facing street view imagery: Insights from a Nanjing case study. *ISPRS International Journal of Geo-Information*, *14*(8), 287

2025 Huang, Y., Zhang, F., Wu, L., & Liu, Y. (2025). Measuring urban physical environments using image deep features. *Cities*, *166*, 106196

2025 Huang, Y., Zhang, F., Li, Y., Wu, L., & Liu, Y. (2025). Intelligent computational representation of urban imagery [In Chinese]. *Geomatics and Information Science of Wuhan University*

2025 Huang, Y., Sanatani, R., Liu, C., Kang, Y., Zhang, F., Liu, Y., Duarte, F., & Ratti, C. (2025). No "true" greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement. *Building and Environment*, *269*, 112395

2023 Huang, Y., Zhang, F., Gao, Y., Tu, W., Duarte, F., Ratti, C., Guo, D., & Liu, Y. (2023). Comprehensive urban space representation with varying numbers of street-level images. *Computers, Environment and Urban Systems*, *106*, 102043

2021 Bian, M., Guo, S., Wang, W., Ouyang, Y., Huang, Y., & Fei, T. (2021). Next-day forecast of Beijing pollen concentration fused with vegetation remote sensing data: Implementing a nonlinear autoregressive neural network model with external input [In Chinese]. *Journal of Geo-Information Science*, *23*(9), 1705–1713

2020 Huang, Y., Fei, T., Kwan, M.-P., Kang, Y., Li, J., Li, Y., Li, X., & Bian, M. (2020). GIS-based emotional computing: A review of quantitative approaches to measure the emotion layer of human–environment relationships. *ISPRS International Journal of Geo-Information*, *9*(9), 551

2020 Huang, Y., Li, J., Wu, G., & Fei, T. (2020). Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus. *Cities*, *102*, 102719

2020 Li, Y., Fei, T., Huang, Y., Li, J., Li, X., Zhang, F., Kang, Y., & Wu, G. (2020). Emotional habitat: Mapping the global geographic distribution of human emotion with physical environmental factors using a species distribution model. *International Journal of Geographical Information Science*, 1–23

2020 Zhang, S., Li, J., Wang, S., Huang, Y., Li, Y., Chen, Y., & Fei, T. (2020). Rapid identification and prediction of cadmium–lead cross-stress of different stress levels in rice canopy based on visible and near-infrared spectroscopy. *Remote Sensing*, *12*(3), 469

2017 Chen, Y., Qi, T., Huang, Y., Wan, Y., Zhao, R., Qi, L., Zhang, C., & Fei, T. (2017). Optimization method of calibration dataset for VIS-NIR spectral inversion model of soil organic matter content [In Chinese]. *Transactions of the Chinese Society of Agricultural Engineering*, *33*(6), 107–114

Conference Proceedings

2026 Huang, Y., Janowicz, K., Karimi, M., Liu, Z., Wang, S., Süß, A., & Fortacz-Lazan, A. (2026). Investigating the multi-scale reproducibility of GeoAI models for urban perception [Accepted, full paper]. *Proceedings of the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*

2026 Huang, Y., Janowicz, K., Karimi, M., Liu, Z., Wang, S., & Süß, A. (2026). Characterizing multidimensional landscape discontinuities at phantom borders with satellite foundation-model embeddings [Accepted, poster paper]. *Proceedings of the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*

2026 Wang, S., Janowicz, K., Benitez Cortes, A., Borderon, M., Huang, Y., Karimi, M., Liu, Z., Süß, A., & Sakdapolrak, P. (2026). Conceptual vagueness as humans and LLMs represent: Evidence from migration distance categorization [Accepted, oral presentation]. *Proceedings of the 16th International Conference on Spatial Information Theory (COSIT 2026)*

Patents

2025 Zhang, F., Huang, Y., & Liu, Y. (n.d.). *Method, device, and storage medium for comprehensive landscape representation based on street imagery* (ZL202510142438.1) [China patent, granted December 16, 2025]

2023 Zhang, F., Huang, Y., & Guo, D. (n.d.-a). *Image processing method, device, product, equipment, and medium* (CN116977690A) [China patent application, published October 31, 2023; patent pending]

2023 Zhang, F., Huang, Y., & Guo, D. (n.d.-b). *Method and related device for identifying regions in space* (CN116958803A) [China patent application, published October 27, 2023; patent pending]

GRANTS

2025–2029 Key Program of the National Natural Science Foundation of China (No. 42430106): Spatio-temporal optimization with geographical artificial intelligence. Contributor.

2024–2026 National Natural Science Foundation of China (No. 42371468): Representing urban physical environment using street view imagery. Contributor.

2022–2025 NSFC Excellent Young Scientists Fund Program (Overseas) (No. 41901321): Geospatial data mining & visual intelligence. Contributor.

2022–2023 International Research Center of Big Data for Sustainable Development Goals (CBAS): Key technologies and applications of big earth data for the SDGs. Contributor.

2021–2022 Tencent Academic Research Fund (No. RAGR20210101): Identifying urban informal settlement using multi-source big geo-data. Key contributor.

2020–2022 National Natural Science Foundation of China (No. 41901321): Understanding the relationship between urban physical environment and urban mobility patterns using street view imagery. Contributor.

2017–2021 National Basic Research Program of China (No. 2017YFB0503602): Geospatial big data mining and spatiotemporal pattern discovery. Contributor.

HONORS AND AWARDS

2025 Postdoc-NeT-AI Fellowship on Explainable AI, German Academic Exchange Service (DAAD), Germany

2022 Visiting Doctoral Research Scholarship, China Scholarship Council (CSC)

2020 "Excellent Graduate Student," Wuhan University

2020 Kwang-Hua Scholarship, Wuhan University

2020 Second-Class Scholarship, Wuhan University

2015 Third-Class Scholarship, Wuhan University

TALKS

Invited Talks

2025 Analysis methods and applications of street view imagery, Nanjing Normal University, China (online)

2025 Intelligent computational representation of urban imagery, GISChat (online)

Conference Talks

2025 Measuring urban physical environments using image deep features, 26th Inter-University Symposium on Asian Megacities, Nanchang, China

2024 Comprehensive urban space representation with varying numbers of street-level images, American Association of Geographers 2024 Annual Meeting, Honolulu, USA

2021 Quantifying the bias in place emotion extracted from photos on social networking sites, 2021 International Graduate Workshop on GeoInformatics, Beijing, China (online)

Conference Posters

2026 No "true" greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement, EGU General Assembly 2026, Vienna, Austria

TEACHING

2026	Geo-Data and Geo-Data Management, University of Vienna (Lecturer, Winter 2026)
2026	Geographic Information Retrieval, University of Vienna (Lecturer, Winter 2026)
2026	Conceptual Modelling and Programming, University of Vienna (Lecturer, Summer 2026)
2025	Spatial Analysis and GIS Applications, Peking University (Teaching Assistant, Spring 2025)

SERVICE

Conference Organization

2026	Co-organizer, 1st International Workshop on Geography According to Foundation Models, AGILE 2026, Tartu, Estonia

Guest Editor

Transactions in Urban Data Science and Technology

Journal Reviewer

ISPRS Journal of Photogrammetry and Remote Sensing

International Journal of Geographical Information Science

Landscape and Urban Planning

Building and Environment

Cities

Computers, Environment and Urban Systems

Sustainable Cities and Society

Environment and Planning B: Urban Analytics and City Science

IEEE Transactions on Human-Machine Systems

Urban Forestry & Urban Greening

Travel Behaviour and Society

Applied Geography

Geography and Sustainability

Scientific Reports

PLOS ONE

Journal of Urban Technology

Computational Urban Science

International Journal of Applied Earth Observation and Geoinformation

International Journal of Digital Earth

Environment International
City and Environment Interactions
International Journal of Disaster Risk Reduction
Journal of Asian Architecture and Building Engineering
Geomatics and Information Science of Wuhan University (in Chinese)
Journal of Transportation Engineering and Information (in Chinese)